MATH 111 Calculus 1

Continuity

Week 2 · Lecture 4 · Stewart §1.8

Dr. Salnikova Chekeres · New Uzbekistan University · Fall 2026

While you wait

Sit with your team. Notebook and pen out: today you write a definition, test it on pictures, and finish with an exit ticket.

Two questions from Lecture 3

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"Why bother with ε and δ if the limit laws compute every limit we meet?"

Because the laws themselves had to be proved, and because "as close as we like" is the only thing we can say when there is no formula. The definition is the court of last appeal; from now on we mostly reap what it gives — starting with continuity.

STAND UP

"The number of people in this hall is a *continuous* function of time."

Stand up if you think this is TRUE.

Stay standing until I say "sit". Then find someone with the other opinion — 60 seconds.

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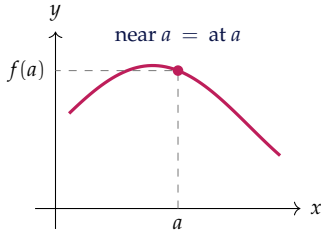
Verdict

FALSE. Every time someone walks in, the number jumps by 1: the graph is a staircase. The *temperature* in the hall, on the other hand, changes continuously — no jumps, no gaps. Continuity means: no jumps, no holes, no blow-ups. Today we say this with one equation.

Continuity at a point

Definition

A function f is continuous at a number a if $\lim_{x \to a} f(x) = f(a)$.

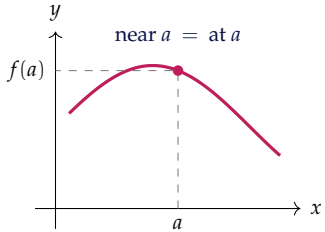


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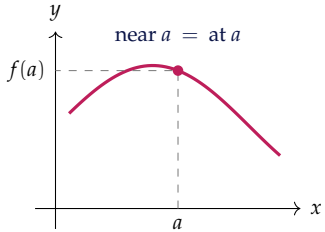
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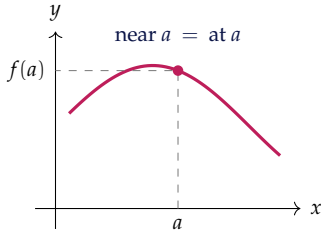
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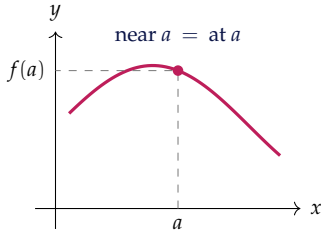
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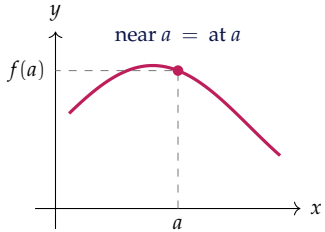
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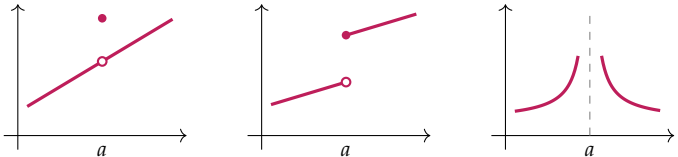


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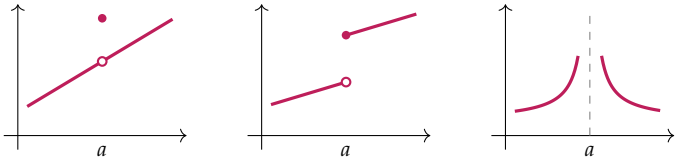
If f is defined near a but not continuous at a: f is discontinuous at a (it has a *discontinuity* at a).

Three ways to fail



removable **jump** **infinite**

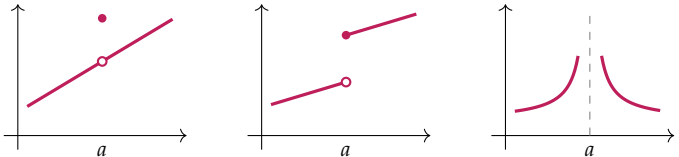
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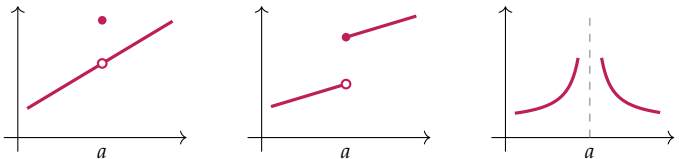


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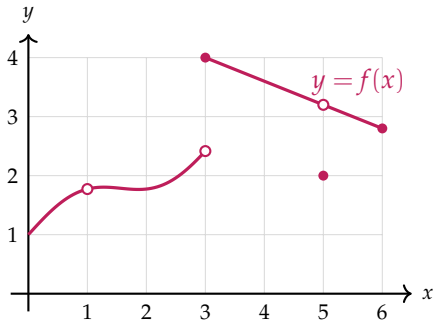
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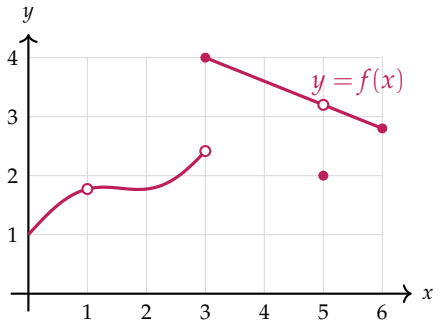
Infinite: $f(x) \to \pm\infty$ on at least one side; the line $x = a$ is a vertical asymptote. A value at a does not help.

Reading discontinuities off a graph



Where is f discontinuous — and why?

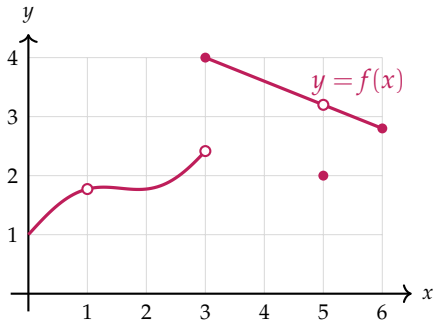
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At 1: the graph has a hole and $f(1)$ is not defined. Condition 1 fails — *removable*.

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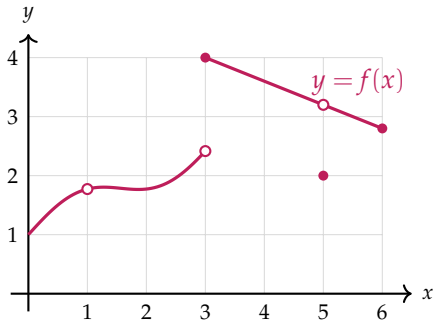


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At 3: $\lim_{x \to 3^-} f(x) \approx 2.4$ but $\lim_{x \to 3^+} f(x) = 4$. The limit does not exist — a *jump*. ($f(3) = 4$ is defined; it does not help.)

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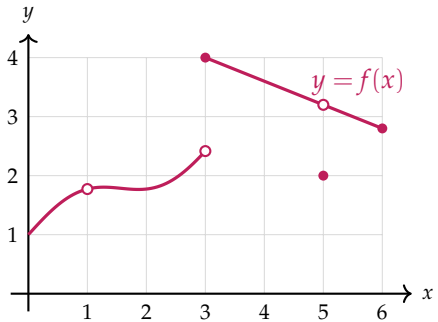
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At 5: $\lim_{x\to 5} f(x) = 3.2$ exists, $f(5) = 2$ is defined — but they are *not equal*. Condition 3 fails — *removable*.

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Everywhere else on $[0, 6]$: continuous. Three failures, three different reasons.

Question 8

Let $f(x) = \begin{cases} \dfrac{x^2-1}{x-1}, & x \neq 1, \\ 3, & x = 1. \end{cases}$ Which statement is true?

A. f is continuous at 1, because $\lim_{x\to 1} f(x)$ exists.
B. f is not continuous at 1; the discontinuity is removable.
C. f is not continuous at 1; the discontinuity is infinite.
D. f is not continuous at 1; the discontinuity is a jump.

In your team

Write your own letter first — then talk. Fingers up: 1 = A, 2 = B, 3 = C, 4 = D. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

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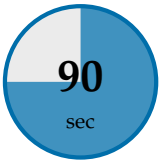


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Hands down. Thank you.

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Who explains?

Hands up — one of you gives the reason; I repeat it aloud for everybody.

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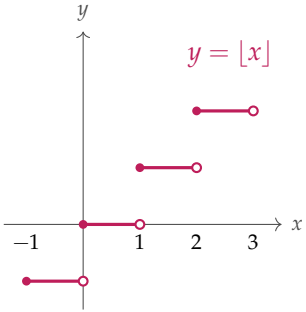
B. For $x \neq 1$, $f(x) = x + 1$, so $\lim_{x\to 1} f(x) = 2$ — but $f(1) = 3$. The limit exists and the value exists; they are not equal. Redefine $f(1) = 2$ and f becomes continuous: removable. (A is the trap: an existing limit is *one* of three conditions, not all of them.)

One-sided continuity

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f is continuous from the right at a if
$\lim_{x \to a^+} f(x) = f(a)$,

and continuous from the left at a if
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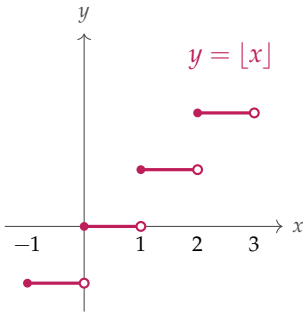
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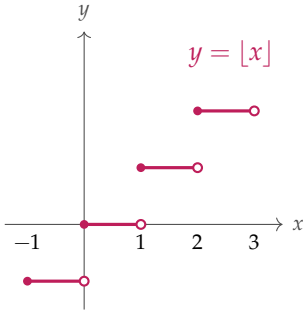
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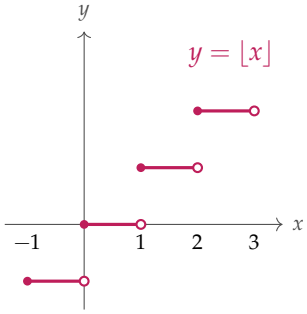
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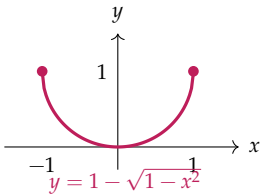
So at a jump a function may still be continuous from *one* side: two-sided continuity is the two one-sided conditions together.



Continuity on an interval — and an example

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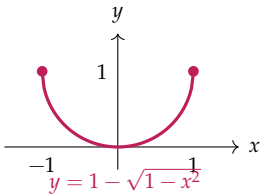
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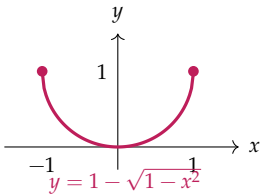


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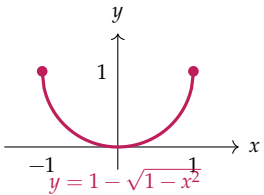
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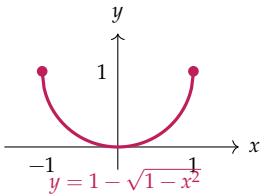
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Hence f is continuous on $[-1, 1]$: one unbroken curve.

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Polynomials, rational functions, root functions and trigonometric functions are continuous at every number in their domains.

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Careful with the words. $f(x) = 1/x$ is continuous — at every number in its domain. At 0 it is not defined, so continuity at 0 is not even a question.

A little history · 1748 and 1821

For Euler (1748) a function was "continuous" if it was given by a *single formula* — so $|x|$, which needs two formulas, counted as discontinuous, and a smooth curve drawn freehand was not a function at all. Bolzano (1817) and Cauchy (1821) replaced formulas by behaviour: continuous means that a small change in x produces a small change in $f(x)$. Today's definition is Cauchy's, written with the limit — and $|x|$ is as continuous as anything.

STAND UP

"The graph of a continuous function can always be drawn without lifting the pen."

Stand up if you think this is TRUE.

Stay standing until I say "sit". Then find someone with the other opinion — 60 seconds.

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Verdict

FALSE as stated. $f(x) = 1/x$ is continuous — at every number in its domain — and its graph has *two* pieces. The pen slogan is true on an interval: continuous on $[a, b]$ means one unbroken curve from $(a, f(a))$ to $(b, f(b))$. Say "continuous on its domain", and name the interval.

Question 9 — error hunt

For which c is $g(x) = \begin{cases} cx + 1, & x \le 2, \\ x^2 - c, & x > 2 \end{cases}$ continuous everywhere? A student writes four lines; exactly one is wrong. Which?

A. Line 1: For continuity at 2 we need $\lim_{x \to 2^-} g(x) = \lim_{x \to 2^+} g(x) = g(2)$.

B. Line 2: $\lim_{x \to 2^-} g(x) = 2c + 1$.

C. Line 3: $\lim_{x \to 2^+} g(x) = c \cdot 2 + 1 = 2c + 1$.

D. Line 4: Therefore g is continuous for every value of c.

In your team

Write your own letter first — then talk. Fingers up: 1 = A, 2 = B, 3 = C, 4 = D. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

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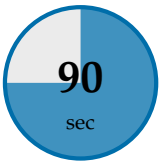
120 sec

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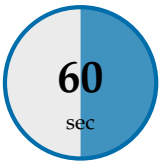
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Hands down. Thank you.

A B C D

Question 9 — error hunt

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Answer

C. Line 3 uses the wrong branch. For $x > 2$ the formula is $x^2 - c$, so $\lim_{x \to 2^+} g(x) = 4 - c$. Continuity at 2 then says $2c + 1 = 4 - c$, i.e. $c = 1$. Line 4 follows correctly from the wrong line 3 — and a conclusion like "every c works" should have felt suspicious.

Making a piecewise function continuous — the method

$$g(x) = \begin{cases} cx + 1, & x \le 2, \\ x^2 - c, & x > 2. \end{cases}$$

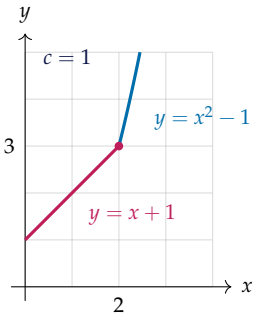
Away from 2 each branch is a polynomial — continuous automatically. Only the seam $x = 2$ needs checking.

Left: $\lim_{x \to 2^-} g(x) = 2c + 1$ (also $g(2) = 2c + 1$).

Right: $\lim_{x \to 2^+} g(x) = 4 - c$.

Continuous at 2 $\iff$ $2c + 1 = 4 - c$ $\iff$ $c = 1$.

Check: with $c = 1$ both branches pass through $(2, 3)$ — the two pieces meet.



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This week's quiz question comes from Questions 8–13

Homework: Stewart §1.8 — read and work through the examples.

EXIT TICKET

Tear a quarter page from your notebook — one each.

1. Your name.
2. One sentence: *"f is continuous at a means …" (in your own words, no formula).*
3. Your muddiest point — the one thing still unclear.
4. One moment today that surprised you.

Give it to the professor or the course assistant as you go. Thank you.